

Contents

The 56-chapter structure shared by the Main book, the Exercise Workbook, and the Answers & Solutions book.

Part I · Getting Your Bearings in Rust

- 1 Why Rust Feels Different
- 2 The Rust Mental Model
- 3 Project Structure and Tooling

Part II · Memory, Ownership, and Data Layout

- 4 Ownership, Borrowing, and Lifetimes
- 5 Pointers, References, and Ownership Inside Structs
- 6 Pointers, References, and Ownership Inside Vectors
- 7 Copying Data vs Cloning Data
- 8 Undefined Behavior and Unsafe Rust
- 9 Smart Pointers and Pinning
- 10 Arrays, Slices, and Vectors
- 11 Hash Maps and Sets
- 12 Matrices and Multidimensional Data
- 13 Arena Allocation and Region-Based Memory

Part III · Designing Types and Abstractions

- 14 Interfaces in Rust: Traits
- 15 OOP Models in Rust
- 16 Domain-Driven Design in Rust
- 17 Refactoring Toward Idiomatic Rust
- 18 Generics Instead of Templates
- 19 Serialization and Data Contracts
- 20 Metaprogramming
- 21 Reflection and Type Introspection

Part IV · Concurrency, Async, and Systems IO

- 22 Multithreading in Rust
- 23 Synchronization Primitives
- 24 Coroutines, Futures, and Async Rust

- 25 Tokio
 - 26 Task Libraries and Parallel Execution
 - 27 IO Tricks and Systems Programming Patterns
-

Part V · Integration, Distributed Systems, and HPC

- 28 C++ Integration
 - 29 JavaScript and C++ Integration for WASM
 - 30 AMQP and Message Brokers
 - 31 Distributed Task Execution
 - 32 MPI and High-Performance Computing
 - 33 Performance-Oriented Rust
 - 34 Memory Profiling
 - 35 Performance Profiling
 - 36 Distributed Tasks Profiling
 - 37 CUDA and GPU Acceleration
 - 38 Merkle Tree Games and Challenges
 - 39 Graph Search Games
 - 40 Matrix Optimization Games
-

Part VI · Engineering Systems for Production

- 41 Error Handling in Large Systems
 - 42 Testing Advanced Rust Systems
 - 43 Observability
 - 44 Packaging and Deployment
 - 45 Capstone: A Distributed Rust System
-

Part VII · Networked Services and Secure Boundaries

- 46 FastAPI-Style Web Apps, Swagger, and OpenAPI Codegen
 - 47 gRPC Services with Protobuf and Service API Codegen
 - 48 WebSockets and Long-Lived Connections
 - 49 HTTPS, TLS, and Secure Service Boundaries
 - 50 libp2p and Peer-to-Peer Rust Systems
-

Part VIII · Frontier Specializations

- 51 Zero-Knowledge Proofs for Rust Engineers
- 52 ZoKrates Workflows and Ethereum Verifiers
- 53 EZKL, Verifiable LLM Inference, and GPU-Aware ZKML

- 54** ONNX Model Inference in Rust
- 55** Smart Contracts in Rust: Solana, Sei, and Beyond
- 56** no-std Rust and Constrained Runtime Derivatives